

Cluster 63227

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Subjects

Research fields: Engineering, Computer science, Mathematics

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Control theory, Electronic engineering, Topology

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: active power filter, control strategy, harmonic detection method, current control, harmonic current

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

0.00% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

122 recent articles in the cluster

122 articles in this cluster were published in the five year period ending in 2023.

Average article is **10.78** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **0.97th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 0.97% of other clusters.

Citation rating: **14.87th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 14.87% of all other articles published in the same year.

33.61% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 116 of 122 articles. 116 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	101	0.12
Russia	7	0.24
India	2	2.29
Indonesia	2	1.00

Top funding countries

Data available for 9 of 122 articles. 9 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	9	0.70

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

An Improved Topology and Control Strategy of Bidirectional PCS

2020: 电力电子技术. 0 citations.

Grid Voltage Sensorless Model Predictive Control for Grid-Connected Inverters

2020: 电工技术学报. 0 citations.

[Computer Simulation on the Basis of SIMINTECH Asynchronous Wind Generator, 2023](#)

2023: Applications and Manufacturing. 2 citations.

Application of Proportional Repetitive Controller in Active Power Filter

2020: 电力电子技术. 0 citations.

Research on Quasi-proportional Resonance Control of LCL Three-phase Four-wire APF

2021: 电力电子技术. 0 citations.

STATCOM Current Control Strategy Based on Feedback Repetitive Control

2020: 电力电子技术. 0 citations.

[Novel soft-start method for three-phase voltage source rectifier based on virtual resistor control](#)

2020: Archives of Electrical Engineering. 1 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

No data available.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

Analysis on AC/DC Harmonic Coupling Characteristics of Converter Based on Harmonic State Space Modeling

2020: 电力系统自动化. 14 citations.

[Atlas de las lenguas del mundo en peligro](#)

2022: Zenodo (CERN European Organization for Nuclear Research). 27 citations.

Comprehensive Power Quality Control for Microgrid Based on APF with Energy Storage

2022: 高压电器. 9 citations.

Grid Voltage Sensorless Model Predictive Control for Grid-Connected Inverters

2020: 电工技术学报. 0 citations.

[Fuzzy and ANFIS Controllers to Improve the Power Quality of Grid Connected PV System with Cascaded Multilevel Inverter](#)

2021: International journal of electrical and electronics research. 17 citations.

Control strategy research and simulation analysis of an independent optical storage microgrid based on voltage stabilizing control

2020: 电力系统保护与控制. 10 citations.

[Improved NLMS-based adaptive denoising method for ECG signals.](#)

2021: Technology and health care : official journal of the European Society for Engineering and Medicine. 8 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 122 out of 122 articles. 62 of these articles came from these sources.

Source	Articles (last five years)
电力电子技术	30
电力系统保护与控制	7
Journal of Physics: Conference Series	5
电机与控制学报	5
电力自动化设备	3
International Conference on Industrial Engineering, Applications and Manufacturing	3
电源学报	3
Energies	2
电网与清洁能源	2
控制工程	2

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 122 of 122 articles. 19 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Nikolai S. Barakin	4	0.06
Andrey A. Kumeyko	3	0.08
王勇	3	0.00
Wang Yong	3	0.00
Nikolay S. Barakin	3	0.67
Yang Jianfeng	2	0.00
刘金昌	2	0.00
Zhang Maosong	2	0.00
张茂松	2	0.00
Wang Qunjing	2	0.00

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 119 of 122 articles. 26 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Kuban State Agrarian University - Russia	6	0.21
Jiangsu University - China	3	0.13
Shanghai Jiao Tong University - China	3	0
兰州交通大学自动化与电气工程学院 - China	3	0
Anhui University - China	3	0
710000	2	0
上海宝准电源科技有限公司 - China	2	0
江苏大学,电气信息工程学院 - China	2	0
710048	2	0
Taiyuan University of Technology - China	2	0

Organization types

This table covers the 61 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 122 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
3.05%	46.56%	6.11%	2.29%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 4 out of 122 articles. 4 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Shanghai Electric (China)	1	0
Commercial Aircraft Corporation of China (China)	1	0.5
Tianjin Energy Investment Group (China)	1	0
China Huadian Corporation (China)	1	2

Top funders

Data available for 10 out of 122 recent articles. 9 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	4	0.70
Funding Project for Professional Leaders of Higher Vocational Colleges in Jiangsu Province	1	0.33
Shandong Province - China	1	0.00
Anhui Province - China	1	1.50
practice and innovation project of Jiangsu Vocational Institute of Architectural Technology in 2019	1	0.33
Shandong University of Technology - China	1	2.00
Science and Technology Project of State Grid Co., Ltd.	1	0.00
Natural Science Foundation of the Anhui Higher Education Institutions of China	1	1.50
Jiangsu University Natural Science Research Project	1	0.33
National Key Ramp;D Program of China	1	0.40

Funder types

This table covers the 6 articles in this cluster from the last five years with data about their funding organization types (out of 122 articles total).

Commercial	Education	Nonprofit	Government
0.00%	33.33%	0.00%	66.67%

Patents and industry

29 patents cite articles in this cluster (57.55th percentile among all clusters)

3.74% of articles in the cluster are cited by patents

1.87% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
STATIC COMPENSATOR (STATCOM) CONTROL METHOD BASED ON MULTI-MODEL FUZZY NEURAL NETWORK PI	2
HIGH-PRESSURE SMART FLEXIBLE TRANSMISSION PARALLEL COMPENSATION EQUIPMENT IN MICROGRID AND LARGE POWER GRID, AND CONTROL METHOD THEREOF	1
POWER EQUIPMENT TEST DEVICE AND SYSTEM AND RETURNED ENERGY ADJUSTMENT METHOD	1
AVERAGE DYNAMIC PHASOR MODEL SUITABLE FOR MULTI-VOLTAGE-CLASS DIRECT-CURRENT POWER DISTRIBUTION SYSTEM	1
DQ CURRENT DETECTION METHOD OF SINGLE-PHASE PULSE RECTIFIER WITHOUT PHASE-LOCKED LOOP	1
HARMONIC RESONANCE SUPPRESSION SYSTEM OF MICROGRID	1
FRACTIONAL ORDER MULTI-CYCLE 6K +/-1 HARMONIC REPETITIVE CONTROLLER DESIGN METHOD AND FRACTIONAL ORDER MULTI-CYCLE 6K +/-1 HARMONIC REPETITIVE CONTROLLER	1
GRID-CONNECTED CONTROL METHOD OF MICRO-GRID BASED ON AUTO-DISTURBANCE-REJECTION TECHNIQUE	1
OVER-CURRENT SUPPRESSION METHOD FOR ENERGY STORAGE CONVERTER	1

Top patent assignees

Patent assignee	Same-cluster citations
STATE FARM (UNITED STATES)	4
UNIVERSITY OF SHANGHAI JIAOTONG	2
SHANGHAI JIAO TONG UNIVERSITY	2
JILIN POWER CO LTD SIPING POWER SUPPLY CO LTD	2
CRRC CORPORATION LTD	1
OCEAN UNIV CHINA	1
DANYANG POWER SUPPLY BRANCH OF STATE GRID JIANGSU ELECTRIC POWER CO LTD	1
国网江苏省电力有限公司镇江供电分公司	1
CRRC (CHINA)	1

Connections

Top citing clusters

No data available.

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
1201	6	LCL Filter, grid-connected inverter, current control, Active Damping Control, grid impedance
65	4	Power Quality Conditioner, Unified Power Quality, Active Power Filter, Shunt Active Power, Controller Based UPQC
4799	2	Direct Power Control, predictive direct power, Active Power Filter, Model Predictive Control, Shunt Active Power
27759	2	Active Power Filter, Shunt Active Power, Power Quality, Power Filters Systems, hybrid shunt active
1972	2	model predictive control, Predictive Current Control, Control Set Model, Set Model Predictive, current control method
14	1	maximum power point, power point tracking, MPPT Techniques, MPPT algorithms, partial shading conditions
3166	1	Adaptive Filter Algorithm, Subband Adaptive Filter, proposed algorithm, Affine Projection Algorithm, Normalized Subband Adaptive
2560	1	energy storage system, proposed control strategy, Microgrid, Distributed Secondary Control, hybrid energy storage
11721	1	Power System, Commutation Failure, HVDC Transmission System, multi-infeed HVDC systems, HVDC Systems
19231	1	proposed rectifier, input current harmonics, power quality, Harmonic Reduction, input line current

Top GitHub repositories

No data available.

The Map of Science runs on bibliometric data from [Clarivate Web of Science](#) and other sources. For details on our sourcing, methodology, and limitations, see the [tool documentation](#).